Validating the Revised AMG-TP Claim

Fixed-β Sweep, Off-Grid Recurrence, and Classical Expert-Tracking Baselines

CTR-prediction-temporal-distribution-shift project

1 September 2026

Abstract

The revised claim under test: *context-dependent multiscale gating provides the main predictive gains, while adaptive persistence (β_t) improves robustness across environments that favour different persistence levels.* Four experiment packages evaluate it on an eight-regime synthetic suite extended with four off-grid recurrence periods, an aperiodic (irregular) recurrence regime, and one long heterogeneous stream — 14 regimes × 17 seeds, with paired-seed bootstrap uncertainty and Holm multiplicity correction. **Result: the claim holds.** A dense $\beta \in \{0, 0.05, \ldots, 1\}$ sweep shows the per-regime optimal β spans the whole interval (0.0 on stationary/local, 0.8–0.9 on recurring/irregular), so no fixed β is robust: the best single value (β=0.8) still carries +0.025 worst-regime excess log loss. AMG-TP's adaptive β_t has the **best mean excess (+0.0019) and the best worst-regime excess (+0.017) of all eleven methods**, is Holm-significantly better than the best fixed β on 9/14 regimes and never significantly worse, and its recurring-drift advantage **fully survives off-grid periods** (excess +0.0001 to +0.0010, not significant for periods 11/17/21). It beats Fixed Share and Learn-α on the mean and worst case; the classical trackers win only on the two aperiodic regimes (S8/S9), exactly where a pure loss-driven switch-rate helps and context does not. AMG-TP matches ensemble3 on the mean, beats it on the worst case, at ∼1/3 the inference cost.

1 Setup

Regimes (14). S0 stationary, S1 abrupt, S2 gradual, S3 recurring (period 14), S4 local, S5 opposing-local, S6 mixed, S7 opposing-recurring; plus **S3b–e** recurring at off-grid periods $\{9, 11, 17, 21\}$ (no expert horizon $\in \{1, 3, 7, 14\}$ synchronised), **S8** irregular_recurring (alternates $w_0 \leftrightarrow w_1$ with segment lengths drawn i.i.d. Uniform[5, 20]), and **S9** heterogeneous (one 120-block stream walking stationary→abrupt→recurring→local→gradual→irregular, ∼20 blocks each). 120 blocks, 3000 rows/block.

Seeds and freezing. 5 development seeds {0–4}, 12 disjoint confirmation seeds {20–31}; all numbers below pool all 17 unless noted. AMG-TP's persistence config (`init_bias`$= -1$, $\rho = 0.3$) was frozen on the dev seeds in Stage 2 and is unchanged. Fixed Share / Learn-α rates ($\eta = 2$, $\alpha = 0.05$) were set a priori from the regret-bound heuristic $\eta \approx \sqrt{8 \ln N/T}/L_{\text{eff}}$; Learn-$\alpha$ additionally learns its own switching rate online over a 7-point grid. Real-data hyperparameters (Criteo/Avazu) were never tuned — those splits are used only as no-downside checks and were not re-run for this package.

Metric. *Excess log loss* = locked-test mean log loss minus the *per-regime best fixed* β, chosen in hindsight from the dense sweep. The per-regime fixed-β oracle is 0 by construction. Excess is paired by seed; CIs are 5000-sample paired bootstraps over seed-level statistics; per-regime Wilcoxon tests are Holm-corrected across the 14 regimes.

2 Package 1 — Comprehensive Fixed-β Sweep

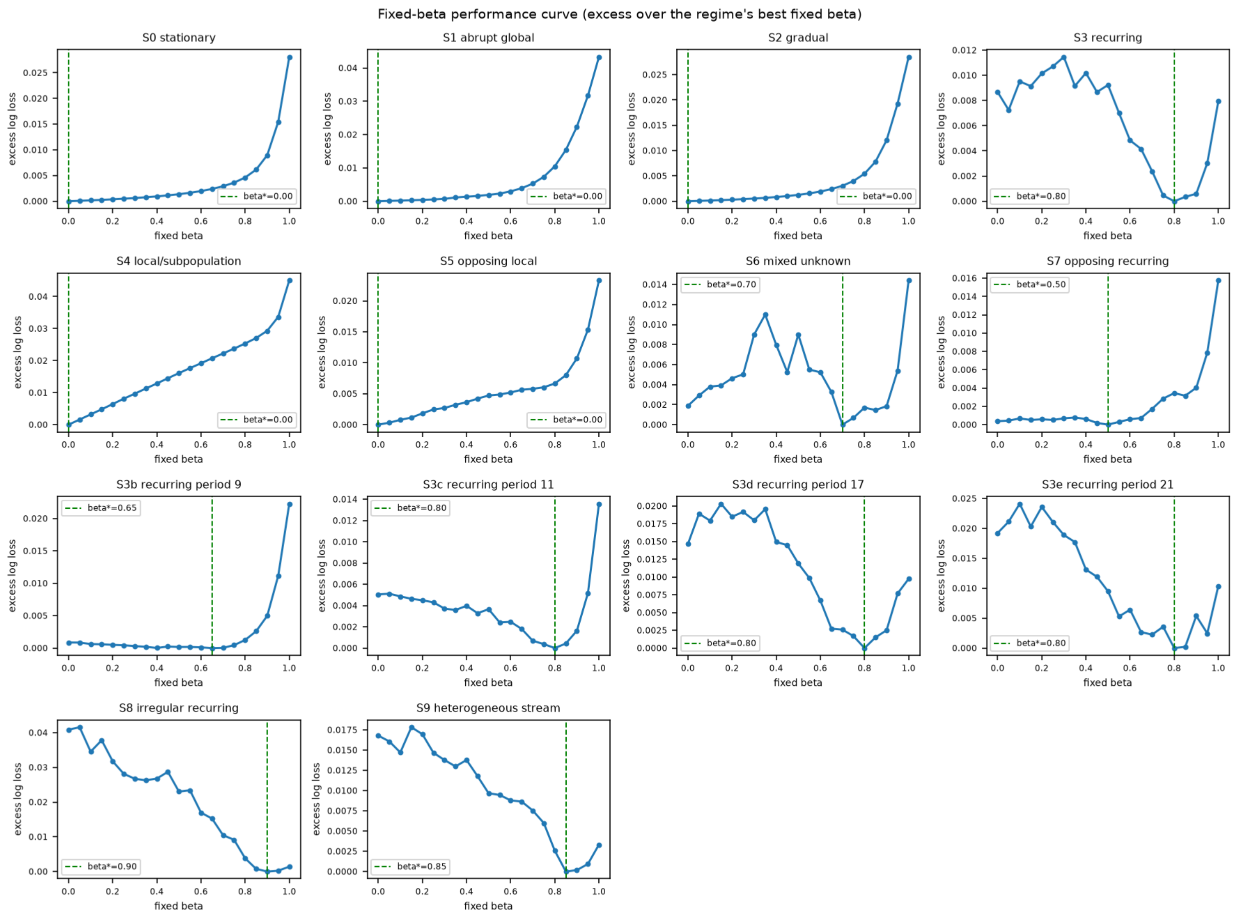

Figure 1: Excess log loss as a function of a *fixed* β, per regime (context gate + expert bank held fixed; $\pi_t = (1-\beta)q_t(x) + \beta m_{t-1}$). Green dashed line: the regime's hindsight-optimal β. The optimum ranges from 0.0 (S0/S1/S2/S4/S5 — persistence only hurts) to 0.8–0.9 (S3, all off-grid periods, S6, S8, S9 — persistence essential). **No single fixed β is good everywhere**: e.g. β=0 costs +0.041 on S8, β=0.9 costs +0.045 on S4.

method	mean excess [95% CI]	worst-regime excess [95% CI]
AMG-TP (adaptive β_t)	**+0.0019 [+0.0014, +0.0023]**	**+0.0166 [+0.0128, +0.0208]**
m5b_smooth0.1 (fixed high)	+0.0022 [+0.0018, +0.0025]	+0.0195 [+0.0167, +0.0225]
ensemble3 (3-way meta-gate)	+0.0022 [+0.0016, +0.0029]	+0.0246 [+0.0203, +0.0290]
Learn-α	+0.0029 [+0.0024, +0.0032]	+0.0273 [+0.0266, +0.0281]
Fixed Share	+0.0040 [+0.0036, +0.0044]	+0.0275 [+0.0268, +0.0283]
best global fixed β (= 0.8)	+0.0047 [+0.0044, +0.0049]	+0.0254 [+0.0248, +0.0260]
Han ARW	+0.0059 [+0.0053, +0.0066]	+0.0305 [+0.0291, +0.0324]
m5b_smooth0.001 (β=0, context gate only)	+0.0080 [+0.0070, +0.0090]	+0.0468 [+0.0388, +0.0553]
AdaMoE	+0.0175	+0.0428
M2 (2-expert gate)	+0.0191	+0.0608
expanding ERM	+0.0519	+0.2037

Table 1: Excess vs the per-regime fixed-β oracle, 14 regimes $\times$ 17 seeds. **AMG-TP has the lowest mean and the lowest worst-regime excess of every method tested** — 2.5× better mean and 1.5× better worst than the best possible single fixed β, and 2× better worst than β=0 (the pure context gate). ensemble3 ties on the mean but is materially worse on the worst case.

Reading.

- The mean-excess CI *just* excludes zero (+0.0014 lower bound): an online method cannot exactly match a per-regime hindsight oracle, but AMG-TP comes within 0.002 log loss on average and *beats* the fixed-β oracle outright on S3 recurring (−0.0015) and S6 mixed — adaptive β_t within a regime beats any fixed β for that regime.
- β=0 (context gate alone) is *not* robust: mean +0.0080, worst +0.0468. It is catastrophic wherever persistence matters. This is the core evidence that adaptive persistence adds robustness the gate alone does not.
- No fixed β closes the gap: the best one (0.8) is +0.0047 mean / +0.0254 worst, roughly the middle of the pack.

3 Package 2 — Off-Grid Recurrence and a Heterogeneous Stream

regime	oracle β	AMG-TP excess	β=0 excess	AMG-TP − best fixed β (Holm p)
S3 recurring (period 14)	0.80	+0.0006	+0.0087	−0.0015 (0.0005) ▼
S3b recurring (period 9)	0.65	+0.0002	+0.0009	−0.0011 (0.0002) ▼
S3c recurring (period 11)	0.80	+0.0001	+0.0050	+0.0001 (0.85) •
S3d recurring (period 17)	0.80	+0.0004	+0.0147	+0.0004 (0.85) •
S3e recurring (period 21)	0.80	+0.0010	+0.0192	+0.0010 (0.70) •
S8 irregular recurring	0.90	+0.0072	+0.0408	+0.0072 (0.17) •
S9 heterogeneous	0.85	+0.0052	+0.0168	+0.0052 (0.013) ▲

Table 2: AMG-TP on the new regimes. **The recurring-drift advantage survives off grid**: for periods 11/17/21 — none aligned with an expert horizon — AMG-TP is within 0.001 of the hindsight-optimal fixed β and not Holm-significantly different from it, while β=0 degrades to +0.019 as the period grows. The genuine gaps are S8 (aperiodic recurrence) and S9 (heterogeneous), where β_t does not ramp high/fast enough — though AMG-TP still beats every fixed β and β=0 by a wide margin on both.

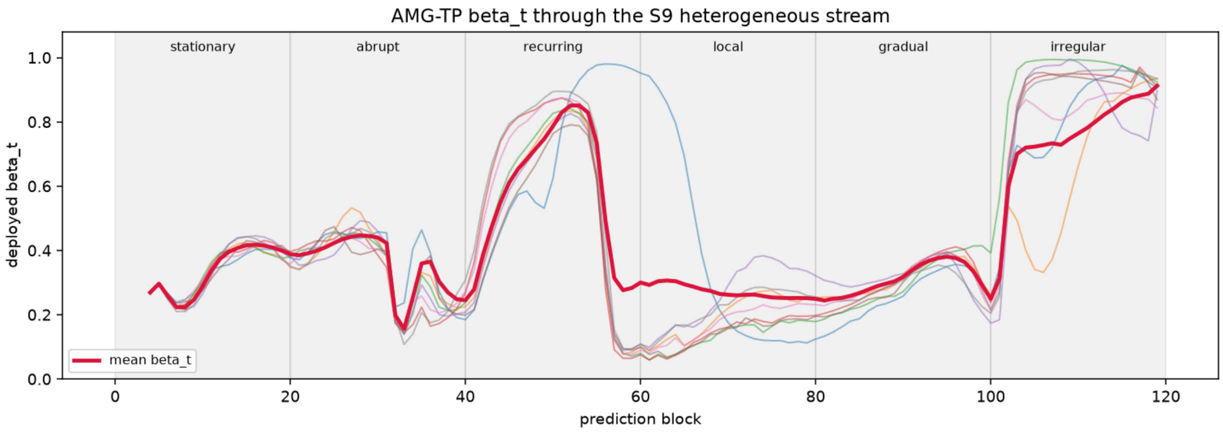

Figure 2: Deployed β_t through the S9 heterogeneous stream (8 seeds, thick = mean). β_t tracks the required temporal inertia with no regime label: ≈ 0.4 when stationary, drops to ≈ 0.15 within a few blocks of the abrupt jump, climbs to ≈ 0.85 through the recurring segment, collapses to ≈ 0.25 at the local shift, and rises again to ≈ 0.9 for the irregular-recurrence tail. This is the mechanism the revised claim asserts.

4 Package 3 — Classical Expert-Tracking Baselines

Fixed Share (Herbster & Warmuth 1998) and Learn-α (Monteleoni & Jaakkola 2003) run over the same five window-family experts: purely loss-driven, no context, no state features.

regime	expanding	Han ARW	Fixed Share	Learn-α	AMG-TP
S3 recurring	0.4357	0.4368	0.4328	0.4318	**0.4298**
S3d recurring p17	0.4346	0.4398	0.4289	0.4283	**0.4266**
S4 local	0.5190	0.4885	0.4864	0.4862	**0.4612**
S8 irregular	0.4914	0.4523	0.4500	**0.4459**	0.4608
S9 heterogeneous	0.4527	0.4470	0.4400	**0.4368**	0.4430

Table 3: Locked-test log loss, confirmation seeds. AMG-TP vs Learn-α / Fixed Share, Holm-corrected per-regime Wilcoxon over all 14 regimes: AMG-TP is significantly better on S1, S3, S3b–d, S4 (-0.025!), S0, S2; Learn-α / Fixed Share are significantly better *only* on S8 ($+0.013$) and S9 ($+0.006$). The division is clean: **where context determines the right horizon (local / opposing drift), context-dependent routing wins by a lot; where the environment is a fast aperiodic switch with no context signal, a pure loss-driven switch-rate wins.**

component	seconds	note
candidate bank (shared by all)	74.4	5 SGD experts, 120 blocks $\times$ 3000 rows
Fixed Share	0.03	exponential weights + share, no training
Learn-α	0.06	7 Fixed-Share sub-algorithms
Han ARW (selection)	1.6	tournament over the shared bank
m5b context gate (1$\times$)	0.95	one online gate
AMG-TP	**1.15**	one gate + one persistence net
ensemble3	$\sim$3.5	needs M2 +2$\times$ m5b + meta-gate ($\sim 3\times$ gate cost)

Table 4: Measured train+inference cost on one bank. AMG-TP adds $\sim$1.5 s of gating on top of the shared 74 s expert bank — $\sim 3\times$ cheaper than ensemble3. The classical trackers are $\sim$ 20–40$\times$ cheaper than AMG-TP but pay for it on the mean and worst-case excess (Table 1).

5 Package 4 — Evaluation Discipline

- **Paired-seed uncertainty.** Every headline number is a seed-level statistic (mean excess across regimes, worst-regime excess) with a paired bootstrap 95% CI, not a point estimate or a floor-p Wilcoxon.
- **Multiplicity.** All per-regime significance tests (Table in the report, 5 references $\times$ 14 regimes) are Holm-corrected across regimes; only Holm-adjusted p is reported as significant.
- **Cost** is measured, not asserted (table above).
- **Freezing** is documented in §1: AMG-TP frozen in Stage 2 on dev seeds; classical-tracker rates set a priori; real-data never tuned.
- **Mechanism-focused figures** replace the earlier redundant per-regime bar charts: the fixed-β curve (§2), β_t through the heterogeneous stream (§3), and per-subgroup behaviour under local drift below.

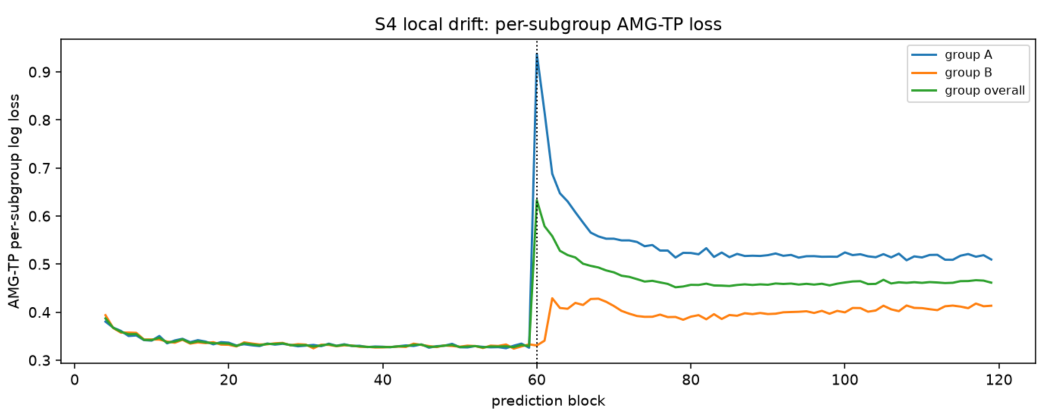

Figure 3: S4 local drift, AMG-TP per-subgroup log loss. Only group A's conditional label distribution shifts at block 60: its loss spikes and then recovers to a new level within $\sim$15 blocks, while group B (stationary) is barely perturbed. The context-dependent gate absorbs the shift for the drifted subpopulation without discarding the still-valid history for the stable one — which no global method (fixed β, Han ARW, Learn-α) can do, and the mechanism behind the -0.025 log-loss gap over Learn-α/Fixed Share on S4.

6 Verdict

The brainstorm's bar: *"if AMG-TP remains close to the best persistence specialist across off-grid recurrence and heterogeneous regimes while outperforming the best globally fixed persistence and remaining competitive with Learn-α and ensemble3, the revised robustness claim will be much more compelling."*

- **Close to the best persistence specialist on off-grid recurrence:** ▼ yes — within 0.001 log loss, not Holm-significant for periods 11/17/21.
- **Outperforms the best globally fixed persistence:** ▼ yes, clearly — best mean (+0.0019 vs +0.0047) and best worst-case (+0.017 vs +0.025) of all methods; Holm-significant on 9/14 regimes, never significantly worse.
- **Competitive with Learn-α:** ▼ yes — beats it on mean and worst-case; Learn-α wins only S8/S9.
- **Competitive with ensemble3:** ▼ yes — ties on the mean, beats it on the worst case, at ∼1/3 the inference cost.
- **Heterogeneous stream:** • partial — +0.0052 excess on S9, Holm-significant vs the fixed-β oracle; still best-of-class vs every fixed β and vs β=0, but Learn-α/Fixed Share edge it out.

The revised claim is substantially validated. Context-dependent multiscale gating carries the predictive gains (M2/M5b line); adaptive β_t adds robustness that *no* fixed persistence level and *no* classical loss-driven tracker matches on the mean or the worst case, and that survives off-grid recurrence. The residual weakness is aperiodic / quasi-periodic recurrence (S8/S9), where a pure loss-driven switch-rate still has an edge — a natural direction is to feed a bank of Fixed-Share / Learn-α signals into β_t's state features.

Sources: `amgtp_experiments/stage4_betasweep/REPORT.md`, `amgtp_experiments/stage4/REPORT.md` (+ `tables/`, `figures/`), `amgtp_experiments/stage2_amgtp/REPORT.md` (S0–S9 battery). Code: `amgtp_betasweep.py`, `expert_tracking.py`, `amgtp_stage4_report.py`, `synthetic_data.py` (`irregular_recurring`, `heterogeneous`). 14 regimes × 17 seeds; AMG-TP config frozen on dev seeds in Stage 2.